

Demonstration of Optical Access Network Failure Management for RAN Control toward Rapid Recovery

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Abstract *We propose RAN control based on failure notifications from the optical access network to shorten failure recovery time for high reliability services. Experiments show that the interruption time can be reduced by 66.9% compared with the conventional method. ©2026 The Author(s)*

Introduction

The sixth-generation (6G) mobile system must meet stringent requirements for new services, such as self-driving cars, telemedicine, and smart cities [1]. Use cases in which artificial intelligence (AI) runs on edge clouds to analyze sensor and video data are now emerging [2]. To realize these services, the access network, which includes not only wireless networks but also optical access networks, must achieve high reliability [3]. Rapid recovery from access network interruptions has become critical. To achieve high reliability, providing a backup path is effective [4]. Recently, the radio access network (RAN) has been structured into a radio unit (RU), a distributed unit (DU), a central unit (CU) to enable flexible functional placement according to network requirements, and an X-haul segment comprising the fronthaul, midhaul, and backhaul [5,6]. To consistently meet the reliability demands of services used during wide-area mobility, physically diverse backup paths must be provided across all X-haul segments. But, in the access network, providing such paths for the large number of geographically distributed cell sites is difficult due to the high attendant costs. In addition, the 6G use of high-frequency bands is expected to significantly increase the number of base stations and antennas [7], making it more difficult to provide backup paths in access networks. Therefore, when an optical access network failure occurs without a backup path within the X-haul, optical access network control alone cannot resolve it, and the UE connection must be switched wirelessly to a different base station. RAN is usually managed by service management and orchestration (SMO) and the RAN intelligent controller (RIC). SMO collects data from DU and CU and recovers failures in these elements or their interfaces within seconds to minutes [8]. However, this SMO recovery time, which ranges from seconds to minutes, is insufficient to meet the stringent real-time communication requirements envisioned for 6G. SMO cannot directly obtain optical access network data from equipment in the X-haul, making accurate identification of the failure condition and location difficult.

In this paper, we propose an optical–wireless

cooperative recovery method in which the optical access network controller (OAN-C) transmits an optical access network failure notification to the near-real-time (Near-RT) RIC via the extended cooperative transport interface (eCTI) [9]. Upon receipt, the Near-RT RIC, operating with a sub-second control cycle [5], immediately instructs the CU to initiate a handover that switches the UE wirelessly to a different base station to bypass the interrupted optical path. This cooperation enables rapid recovery even when a failure occurs in a single-path X-haul segment. To demonstrate the feasibility of the proposed method, we implement failure notification function on the OAN-C and a notification-triggered handover instruction function on the RIC. We also install open-source software (OSS)-based RAN functions and a fifth-generation core network (5GC) on several general-purpose servers and use an optical switch as the optical access network equipment.

Optical Access Network Failure Notification to Near-RT RIC for Fast Handover Initiation

The proposed rapid failure recovery method has the OAN-C transmit an optical network failure notification to the Near-RT RIC after a network interruption. The Near-RT RIC then commands an immediate handover, thereby improving access-network reliability. Figure 1 overviews the proposed method given an optical access network interruption. In the example shown in this figure, the DU is deployed at the cell site, and the midhaul segment is assumed to correspond to the optical access network. In the proposed method, the OAN-C and Near-RT RIC are directly interconnected via eCTI without involving SMO, enabling fast information exchange between the controllers. Near-RT RIC determines the backup DU in advance to reduce recovery processing overhead after failure occurrence. To realize rapid recovery, the OAN-C employs three basic functions: collection, failure detection, and failure notification. The collection function periodically collects the data required for failure detection, such as the received optical power strength, from the optical access network equipment. The failure detection function then flags an interruption when a monitored optical parameter crosses a

predefined threshold. Unlike the conventional method, which determines failures at SMO based on data collected from CU and DU, in the proposed method, the OAN-C determines failures based on data directly acquired from the optical access network equipment, enabling faster detection of X-haul failures. In the proposed method, immediately upon detecting an interruption in the X-haul, the OAN-C transmits a failure notification to the Near-RT RIC, which supports fast control as its control cycle is less than 1 s. The Near-RT RIC remains in a listening state to capture failure notifications from the OAN-C at any time. When the Near-RT RIC receives a failure notification, the radio control xApp, which consumes and/or produces Near-RT RIC services [5], immediately transmits a handover command to the CU, which then sends commands to the DUs to initiate the handover. This direct chain of rapid detection and handover greatly reduces network interruption time.

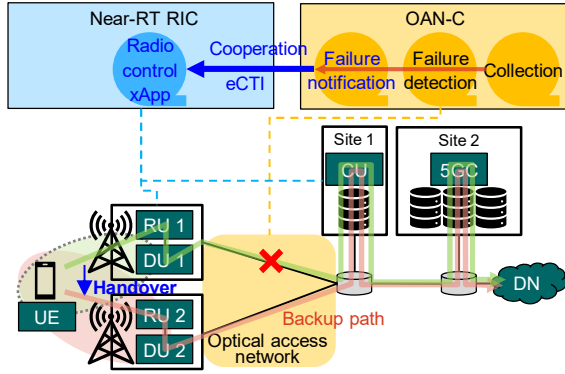


Fig. 1: Overview of the proposed method given an optical access network failure.

Experimental Setup

We demonstrate the feasibility of the proposed fast handover control using an optical switch, which measures optical parameters, OSS-based CU/DU software (Open Air Interface: OAI [10]), and RIC (FlexRIC [11]). Figure 2 shows the experimental setup used to validate our proposal and to evaluate its performance in practice. The OAN-C prototype collected the received optical power strength at each port of the optical switch

by using the standard commands for programmable instruments (SCPI) protocol. We implemented the OAN-C, which transmits a Representational State Transfer (REST) message to the FlexRIC via eCTI upon detecting an interruption. We installed FlexRIC and implemented an xApp that remains in a listening state to capture failure notification REST messages and transmits a handover command to the CU upon receipt of each such message. FlexRIC connected to the CU server and the DU servers. UE, RU, and DU functions were distributed across three servers (UE on server 1; RU 1/DU 1 on server 2; RU 2/DU 2 on server 3), with a wired link via switching hub emulating the UE–RU connections for handover experiments. Server 4 supported the CU, 5GC, and data network (DN). The DU and the CU were connected via a switching hub and an optical switch used for optical power measurement. Path 1 ran from the UE to the DN via DU 1; Path 2 was preconfigured via DU 2 as a backup. To trigger a midhaul interruption, we turned off the media converter (MC) between the optical switch and the DU 1 server. During the midhaul interruption, DU 1 could not receive handover commands from the CU, so an OpenFlow switch was placed to transmit them over a physically separate path. The UE sent ping requests to the DN through the GPRS Tunneling Protocol for the user plane (GTP-U) tunnel at 100 ms intervals, and the resulting ping replies were captured at a capture point A to estimate the network interruption time.

Figure 3 shows the experimental sequence of the proposed method. The OAN-C collected the received optical power strength from the optical switch at 30 ms intervals. When the received optical power strength was below the threshold, the OAN-C transmitted a failure notification to FlexRIC via eCTI. Upon receiving the failure notification, the xApp in FlexRIC transmitted a handover command to CU. The handover process then started, and CU sent the UE context setup request to DU 1 and the UE context modification request to DU 2 [12]. Handover execution finished after UE context release. As the conventional method in this experiment, FlexRIC collects

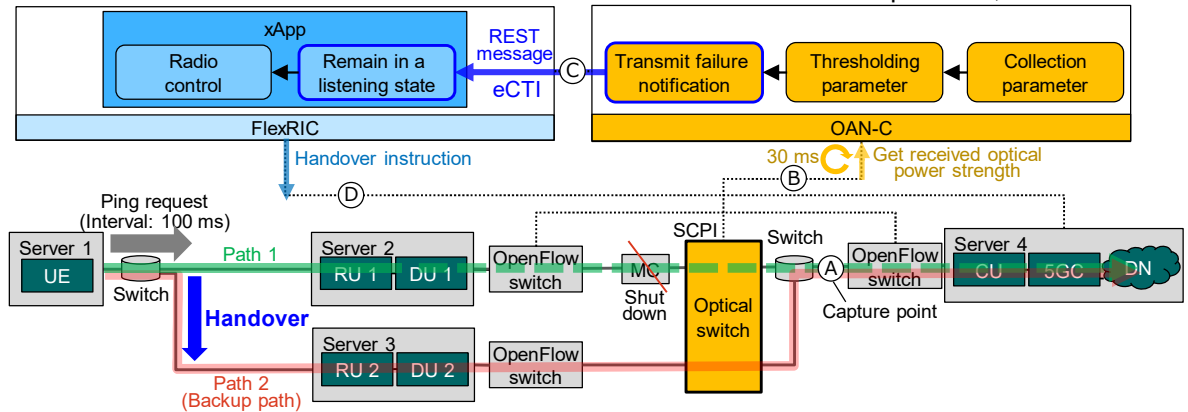


Fig. 2: Experimental setup to evaluate the proposed method.

the UE downlink throughput from the DU 1 at 1 s intervals to emulate SMO data collection. If throughput falls below the threshold, xApp transmits the handover command to the CU. To evaluate the interruption time and each process time in milliseconds, a traffic analyzer captured collection packets at point B, failure notification packets at point C, and handover instruction packets at point D for analysis, as indicated in Fig. 2.

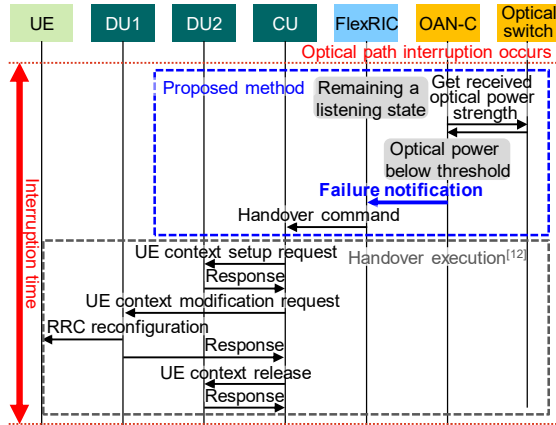


Fig. 3: Sequence of the proposed method in the experiment.

Experimental Results

Figure 4 shows the ping replies versus elapsed time for the trial with the shortest interruption among 10 trials. Figs. 4(a) and 4(b) show the conventional and proposed methods, respectively. Fig. 4(a) shows that, with the conventional method, interruption arises at an elapsed time of 0, and handover finished at 1.63 s; failure detection and handover instruction to the CU by the xApp took 1.16 s, while handover took 0.47 s. The conventional method collected downlink throughput at 1 s intervals in the xApp, which forced the failure detection time to often exceed 1 s, depending on the timing of the collection event. On the other hand, Fig. 4(b) shows that the proposed method completed handover within 0.54 s; failure detection, transmitting a notification from OAN-C to the Flex RIC, and xApp handover instruction to the CU took 0.057 s, while handover execution took 0.49 s. The proposed method directly collects data from the optical switch, reducing the detection time and enabling faster handover initiation by forwarding notifications to the Flex RIC. As a result, the proposed method reduced the interruption time by 66.9% compared to the conventional method.

Figures 5(a) and 5(b) show the average and maximum interruption times across 10 trials for the conventional and proposed methods. The proposed method reduced the average failure detection time from 1.50 s in the conventional method to 0.11 s. The proposed method needs a notification from OAN-C to the Flex RIC, yet the

average transmitting notification time was 7 ms, which remained small relative to the failure detection time. Compared with the conventional method, the proposed method reduced the average time from failure occurrence to handover instruction to CU by 92.3%. Furthermore, the proposed method cut the time required for network interruption detection and notification to the FlexRIC to 0.19 s at most, indicating a substantial reduction in the time required to complete handover.

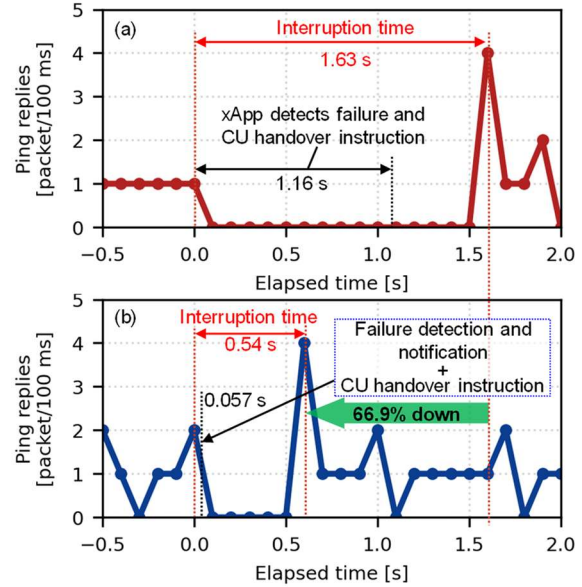


Fig. 4: Ping replies under minimum interruption time:(a) conventional method, (b) proposed method.

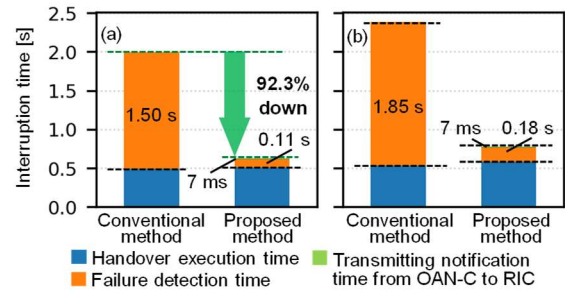


Fig. 5: Interruption time:(a) average time, (b) maximum time.

Conclusion

This paper proposed a rapid network recovery method in which OAN-C transmits a failure notification to the Near-RT RIC to initiate handover. An experiment using an OSS-based CU/DU and an optical switch showed that the proposed method, which implements a failure notification function on the OAN-C and a notification-triggered handover instruction function on the xApp, reduces interruption time by 66.9% and average time from failure occurrence to handover instruction to CU by 92.3% compared to the conventional method, which performs failure detection based on data acquired by FlexRIC. This study will contribute to the realization of highly reliable network services.

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